

Problem Statement: Intelligent Document Processing (IDP) for Insurance Claims

1. Business Context

Insurance companies process thousands of claims daily through multiple input channels such as emails, mobile uploads, and scanned physical documents. These documents contain structured, semi-structured, and unstructured data including handwritten notes, images, and forms. Current systems rely heavily on manual data entry, leading to inefficiencies, delays, and increased costs.

2. Problem Definition

The organization lacks a scalable, automated, intelligent system to process multi-format insurance documents. Key Challenges: - Fragmented ingestion across channels - Manual classification of documents - Inefficient extraction from low-quality scans - Validation bottlenecks - Delayed downstream integration

3. Impact of the Problem

- Increased time-to-settle (days instead of minutes) - High operational costs - Poor customer experience - Increased error rates and compliance risks

4. Desired Future State

A multi-modal AI-powered system that automates ingestion, classification, extraction, validation, and integration with claims systems.

5. Key Requirements

Functional: - Multi-channel ingestion - AI-based classification - Data extraction - Real-time validation - Workflow automation
Non-Functional: - High accuracy (>95%) - Low latency - Scalability - Compliance - High availability

7. Core Insight

Success depends on end-to-end architectural cohesion across data, model, and application layers.

8. Final Summary

Insurance claims processing is hindered by manual and error-prone workflows. A scalable AI-driven IDP solution is required to automate document processing and improve speed, accuracy, and cost efficiency.

6. Success Metrics

Metric	Target Outcome
Processing Speed	Reduce time from days to minutes
Accuracy	Increase STP rate
Cost Efficiency	Reduce manual costs
Customer Experience	Faster settlements